

19-206-0711 ENERGY MANAGEMENT AND CONSERVATION

Module – 2

Evaluation of Thermal Performance

Sizing Air-Conditioning System



- Concepts and fundamentals of air conditioner sizing is based on heat gain, and/or losses in a building.
- We need to remove the amount of heat gain - if it is hot outside.
- Similarly, we need to add in the heat loss from space - if outside temperature is cold.
- In short, heat gain and loss, must be equally balanced by heat removal, and addition, to get the desired room comfort that we want.
- The heat gain or heat loss through a building depends on:
 - The temperature difference between outside temperature and our desired temperature.
 - The type of construction and the amount of insulation is in the ceiling and walls.
 - How much shade is on the building's windows, walls, and roof?

Sizing Air-Conditioning System



- The surface area of the walls.
- Infiltration – How much air leaks into indoor space from the outside?
- The occupants.
- Activities and other equipment within a building. Cooking? Hot bath? Gymnasium?
- Amount of lighting in the room. High efficiency lighting fixtures generate less heat.
- Appliances – How much heat the appliances generate? Number of power equipment such as oven, washing machine, computers, TV inside the space; all contribute to heat.
- The air conditioner's efficiency, performance, durability, and cost depend on matching its size to the above factors.

Sizing Air-Conditioning System



- Rules-of-thumb methods for sizing
 - Many designers use a simple square foot method for sizing the air-conditioners.
 - The most common rule of thumb is to use "1 ton for every 500 square feet of floor area".
 - Such a method is useful in preliminary estimation of the equipment size.
 - The main drawback of rules-of-thumb methods is the presumption that the building design will not make any difference.
 - Thus the rules for a badly designed building are typically the same as for a good design.
- It is important to use the correct procedure for estimating heat gain or heat loss.
- Two groups – the Air Conditioning Contractors of America (ACCA) and the American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE) – publish calculation procedures for sizing central air conditioners.
- Reputable air conditioning contractors will use one of these procedures, often performed with the aid of a computer, to size your new central air conditioner.

Heating Load V/s Cooling Load Calculations



- Heating load estimation
 - Heating load calculations are carried out to estimate the heat loss from the building in winter so as to arrive at required heating capacities.
 - Normally during winter months the peak heating load occurs before sunrise and the outdoor conditions do not vary significantly throughout the winter season.
 - In addition, internal heat sources such as occupants or appliances are beneficial as they compensate some of the heat losses.
 - As a result, normally, the heat load calculations are carried out assuming **steady state** conditions (no solar radiation and steady outdoor conditions) and neglecting internal heat sources.
 - This is a simple but conservative approach that leads to slight overestimation of the heating capacity.
 - For more accurate estimation of heating loads, one has to take into account the thermal capacity of the walls and internal heat sources, which makes the problem more complicated.

Heating Load V/s Cooling Load Calculations



- Cooling load estimation
 - For estimating cooling loads, one has to consider the **unsteady state** processes, as the peak cooling load occurs during the day time and the outside conditions also vary significantly throughout the day due to solar radiation.
 - In addition, all internal sources add on to the cooling loads and neglecting them would lead to underestimation of the required cooling capacity and the possibility of not being able to maintain the required indoor conditions.
 - Thus cooling load calculations are inherently more complicated.
 - In determining the heating load, credit for solar heat gain or internal heat gains is usually NOT included and the thermal storage effects of building structure are generally ignored.
 - In cooling load calculations, the thermal storage characteristics of the building play a vital role because the time at which the space may realize the heat gain as a cooling load will be considerably offset from the time the heat started to flow.

Heat Flow Rates



- In air-conditioning design, four related heat flow rates, each of which varies with time, must be differentiated:
 - Space heat gain – How much heat (energy) is entering the space?
 - Space cooling load – How much energy must be removed from the space to keep temperature and relative humidity constant?
 - Space heat extraction – How much energy is the HVAC removing from the space?
 - Cooling load (coil) – How much energy is removed by the cooling coil serving various spaces plus any loads external to the spaces such as duct heat gain, duct leakage, fan heat and outdoor makeup air?

Space Heat Gain



- This instantaneous rate of heat gain is the rate at which heat enters into and/or is generated within a space at a given instant.
- Heat gain is classified by:
 - The manner in which it enters the space
 - Solar radiation through transparent surfaces such as windows
 - Heat conduction through exterior walls and roofs
 - Heat conduction through interior partitions, ceilings and floors
 - Heat generated within the space by occupants, lights, appliances, equipment and processes
 - Loads as a result of ventilation and infiltration of outdoor air
 - Other miscellaneous heat gains
 - Type of heat gain – Whether it is a sensible or latent gain
 - Sensible heat - Heat which a substance absorbs, and while its temperature goes up, the substance does not change state.

Space Heat Gain



- Sensible heat gain is directly added to the conditioned space by conduction, convection, and/or radiation.
- Note that the sensible heat gain entering the conditioned space does not equal the sensible cooling load during the same time interval because of the stored heat in the building envelope.
- Only the convective heat becomes cooling load instantaneously.
- Sensible heat load is total of
 - » Heat transmitted thru floors, ceilings, walls
 - » Occupant's body heat
 - » Appliance & Light heat
 - » Solar Heat gain thru glass
 - » Infiltration of outside air
 - » Air introduced by Ventilation
- Latent Heat Loads – Latent heat gain occurs when moisture is added to the space either from internal sources (e.g. vapor emitted by occupants and equipment) or from outdoor air as a result of infiltration or ventilation to maintain proper indoor air quality.

Space Heat Gain

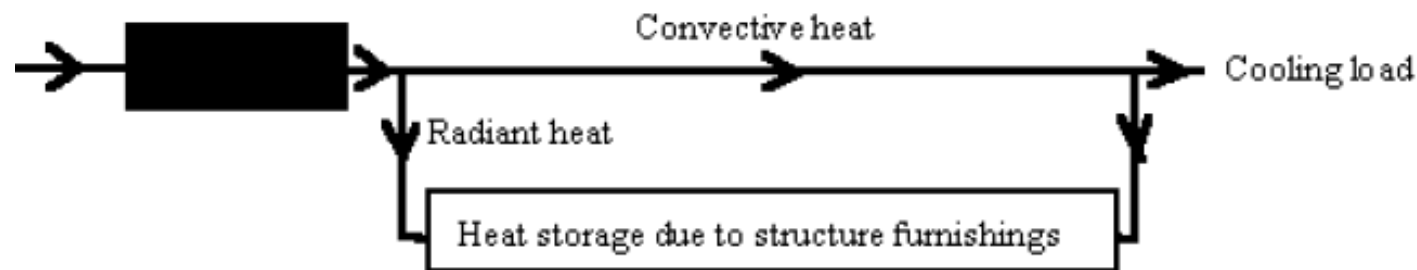


- Latent heat load is total of
 - » Moisture-laden outside air from Infiltration & Ventilation
 - » Occupant Respiration & Activities
 - » Moisture from Equipment & Appliances
- To maintain a constant humidity ratio, water vapor must condense on cooling apparatus at a rate equal to its rate of addition into the space. This process is called dehumidification and is very energy intensive, for instance, removing 1 kg of humidity requires approximately 0.7 kWh of energy.

Space Heat Gain V/s Cooling Load (Heat Storage Effect)

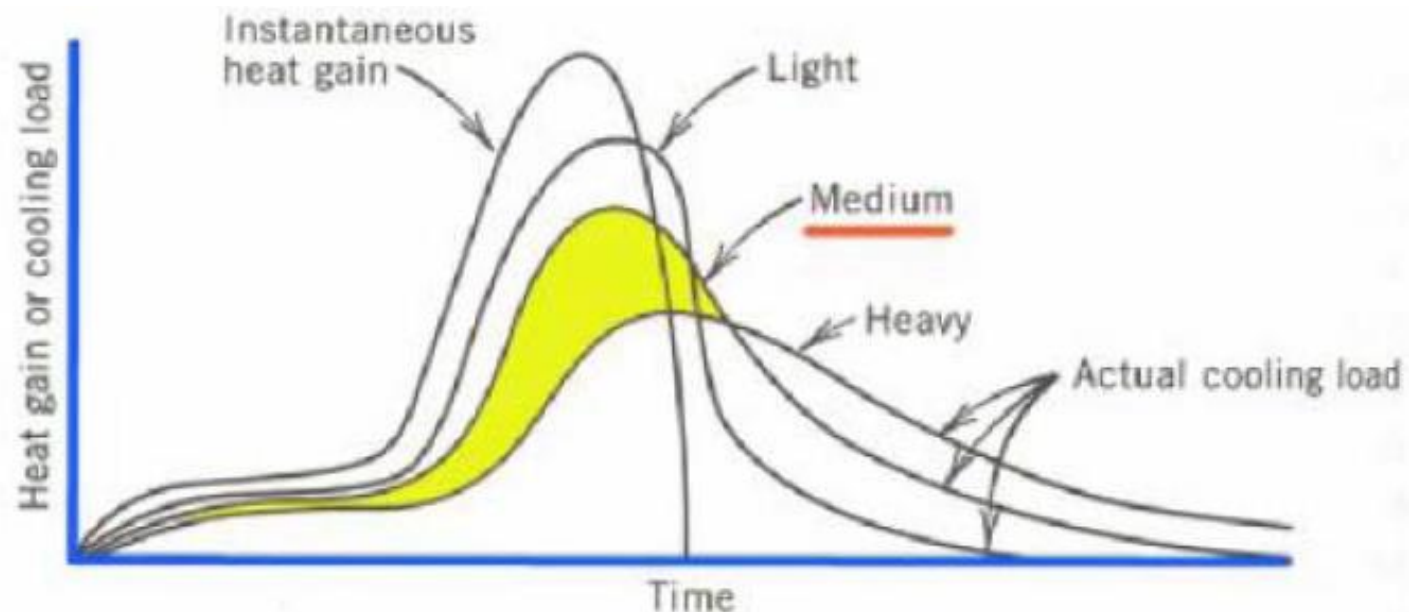


- The heat received from the heat sources (conduction, convection, solar radiation, lightning, people, equipment, etc...) does not go immediately to heating the room air.
- Only some portion of it is absorbed by the air in the conditioned space instantaneously leading to a minute change in its temperature.
- Most of the radiation heat especially from sun, lighting, people is first absorbed by the internal surfaces, which include ceiling, floor, internal walls, furniture etc.
- Due to the large but finite thermal capacity of the roof, floor, walls etc., their temperature increases slowly due to absorption of radiant heat.
- The radiant portion introduces a time lag and also a decrement factor depending upon the dynamic characteristics of the surfaces.
- Due to the time lag, the effect of radiation will be felt even when the source of radiation is removed.



Space Heat Gain V/s Cooling Load (Heat Storage Effect)

- Differences between instantaneous heat gain and cooling load is due to heat storage affect.
- The relation between heat gain and cooling load and the effect of the mass of the structure (light, medium & heavy) is shown below.
- It is evident that, there is a delay in the peak heat, especially for heavy construction.

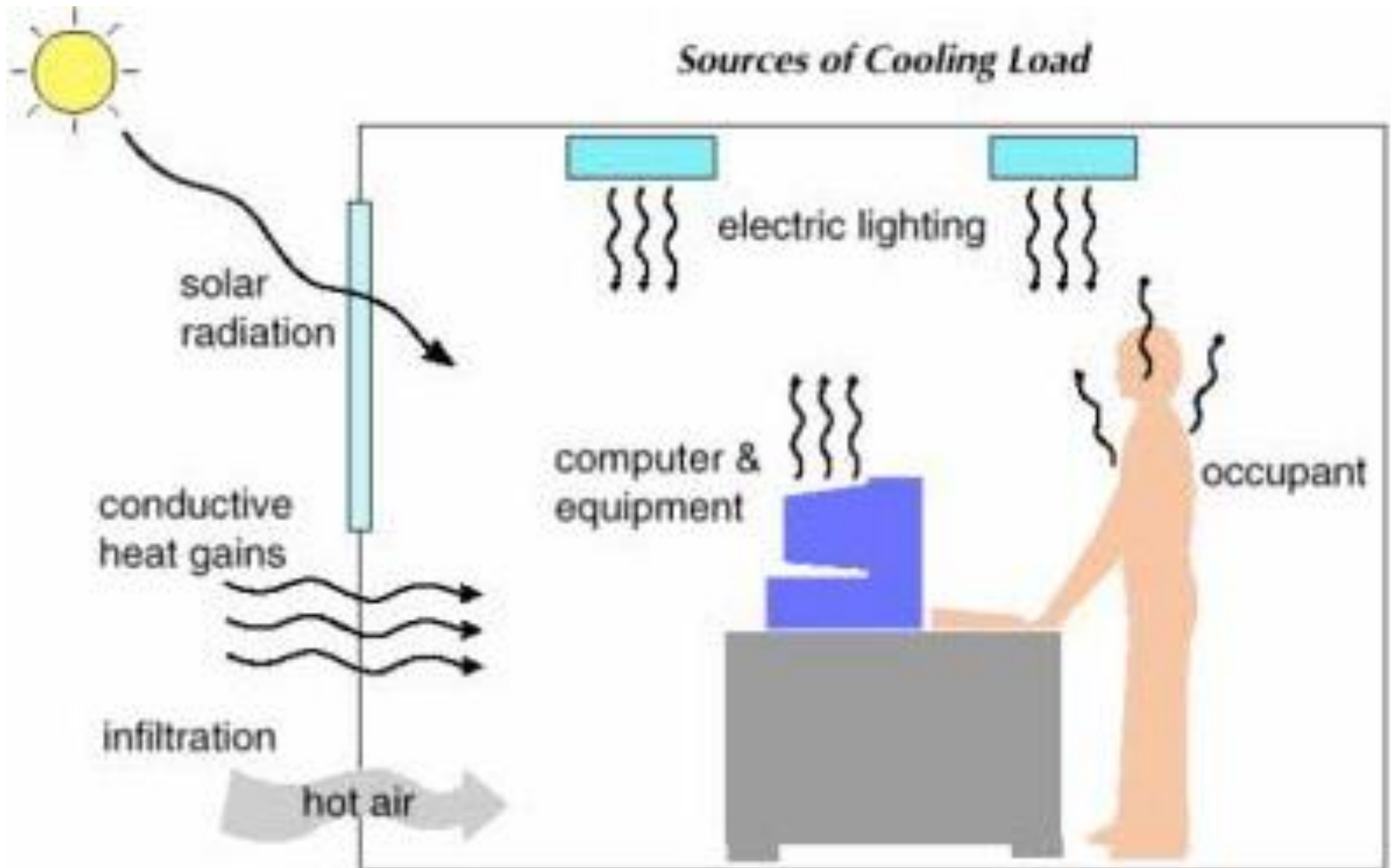


Space Cooling V/s Cooling Load (Coil)



- Space cooling is the rate at which heat must be removed from the spaces to maintain air temperature at a constant value.
- Cooling load, on the other hand, is the rate at which energy is removed at the cooling coil that serves one or more conditioned spaces in any central air conditioning system.
- Cooling load is equal to the instantaneous sum of the space cooling loads for all spaces served by the system plus any additional load imposed on the system external to the conditioned spaces items such as fan energy, duct heat gain, duct leakage, heat extraction lighting systems, etc..

Components of Cooling Load



Components of Cooling Load



- Sources of cooling load
 - The total building cooling load consists of heat transferred through the building envelope (walls, roof, floor, windows, doors etc.) and heat generated by occupants, equipment, and lights.
 - The load due to heat transfer through the envelope is called as external load, while all other loads are called as internal loads.
 - The percentage of external versus internal load varies with building type, site climate, and building design.
 - The total cooling load on any building consists of both sensible as well as latent load components.
 - The sensible load affects the dry bulb temperature, while the latent load affects the moisture content of the conditioned space.
- Buildings may be classified as
 - externally loaded and
 - internally loaded.

Components of Cooling Load



- Externally loaded buildings
 - In externally loaded buildings the cooling load on the building is mainly due to heat transfer between the surroundings and the internal conditioned space.
 - Since the surrounding conditions are highly variable in any given day, the cooling load of an externally loaded building varies widely.
- Internally loaded buildings
 - In internally loaded buildings the cooling load is mainly due to internal heat generating sources such as occupants, lights or appliances.
 - In general the heat generation due to internal heat sources may remain fairly constant, and since the heat transfer from the variable surroundings is much less compared to the internal heat sources, the cooling load of an internally loaded building remains fairly constant.
- Obviously from energy efficiency and economics points of view, the system design strategy for an externally loaded building should be different from an internally loaded building.

Cooling Load Calculation Method



- For a thorough calculation of the zones and whole-building loads, one of the following three methods should be employed:
 - Transfer Function Method (TFM)
 - This is the most complex of the methods proposed by ASHRAE and requires the use of a computer program or advanced spreadsheet.
 - Cooling Load Temperature Differential/Cooling Load Factors (CLTD/CLF)
 - This method is derived from the TFM method and uses tabulated data to simplify the calculation process.
 - The method can be fairly easily transferred into simple spreadsheet programs but has some limitations due to the use of tabulated data.
 - Total Equivalent Temperature Differential/Time-Averaging (TETD/TA)
 - This was the preferred method for hand or simple spreadsheet calculation before the introduction of the CLTD/CLF method.

Cooling Load Methodology – Considerations & Assumptions



- Design cooling load takes into account all the loads experienced by a building under a specific set of assumed conditions.
- The assumptions behind design cooling load are as follows:
 - Weather conditions are selected from a long-term statistical database. The conditions will not necessarily represent any actual year, but are representative of the location of the building. ASHRAE has tabulated such data.
 - The solar loads on the building are assumed to be those that would occur on a clear day in the month chosen for the calculations.
 - The building occupancy is assumed to be at full design capacity.
 - The ventilation rates are either assumed on air changes or based on maximum occupancy expected.
 - All building equipment and appliances are considered to be operating at a reasonably representative capacity.
 - Lights and appliances are assumed to be operating as expected for a typical day of design occupancy.

Cooling Load Methodology – Considerations & Assumptions



- Latent as well as sensible loads are considered.
- Heat flow is analyzed assuming dynamic conditions, which means that heat storage in building envelope and interior materials is considered.
- The latent heat gain is assumed to become cooling load instantly, whereas the sensible heat gain is partially delayed depending on the characteristics of the conditioned space. According to the ASHRAE regulations, the sensible heat gain from people is assumed 30% convection (instant cooling load) and 70% radiative (delayed portion).
- Peak load calculations evaluate the maximum load to size and select the refrigeration equipment. The energy analysis program compares the total energy use in a certain period with various alternatives in order to determine the optimum one.
- Space (zone) cooling load is used to calculate the supply volume flow rate and to determine the size of the air system, ducts, terminals, and diffusers. The coil load is used to determine the size of the cooling coil and the refrigeration system. Space cooling load is a component of the cooling coil load.
- The heat transfer due to ventilation is not a load on the building but a load on the system.

Calculation of Conductive Heat Loss



- A concrete wall has a surface area of 30 m^2 and is 0.30 m thick. It separates warm room air from cold ambient air. The inner surface of the wall is known to be at a temperature of 25°C , while the outer surface is at -15°C . The thermal conductivity of the concrete is $1.0 \text{ Wm}^{-1}\text{K}^{-1}$. What is the heat loss through the wall?

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The driving force is the temperature difference, i.e. $\Delta T = 25 - (-15) = 40^\circ\text{C}$

The heat flux, $Q/A = k \frac{dT}{dx}$ by Fourier's law

$$Q/A \int dx = k \int dT \Rightarrow Q/A \times \Delta x = k \times \Delta T$$

$$\text{Therefore, } Q = \frac{k}{\Delta x} \times A \times \Delta T = \frac{1}{0.3} \times 30 \times 40$$

$$= 4000 \text{ W} = \underline{\underline{4 \text{ kW}}}$$

CLTD/SCL/CLF Method of Load Calculation



- CLTD (cooling load temperature difference), SCL (solar cooling load factor), and CLF (cooling load factor): all include the effect of
 - (1) time-lag in conductive heat gain through opaque exterior surfaces and
 - (2) time delay by thermal storage in converting radiant heat gain to cooling load.
- This approach allows cooling load to be calculated manually by use of simple multiplication factors.
 - CLTD is a theoretical temperature difference that accounts for the combined effects of inside and outside air temp difference, daily temp range, solar radiation and heat storage in the construction assembly/building mass. It is affected by orientation, tilt, month, day, hour, latitude, etc. CLTD factors are used for adjustment to conductive heat gains from walls, roof, floor and glass.
 - CLF accounts for the fact that all the radiant energy that enters the conditioned space at a particular time does not become a part of the cooling load instantly. The CLF values for various surfaces have been calculated as functions of solar time and orientation and are available in the form of tables in ASHRAE Handbooks. CLF factors are used for adjustment to heat gains from internal loads such as lights, occupancy, power appliances.
 - SCL factors are used for adjustment to transmission heat gains from glass.

External Cooling Load by Roof Heat Transfer



- $Q = U * A * (CLTD)$
- Q = Cooling load, W
- U = Coefficient of heat transfer for roof, $W/m^2.^\circ K$
- A = area of roof, m^2
- $CLTD$ = cooling load temperature difference $^\circ K$.

External Cooling Load by Wall Heat Transfer

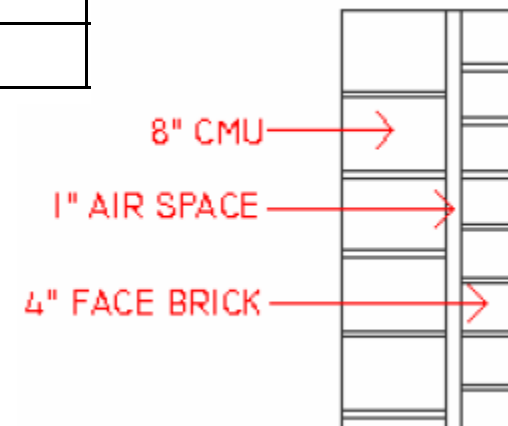


- $Q = U * A * (CLTD)$
- Q = Cooling load, W
- U = Coefficient of heat transfer for wall, $W/m^2.^\circ K$
- A = area of wall, m^2
- $CLTD$ = cooling load temperature difference $^\circ K$.

Estimation of Coefficient of Heat Transfer

- The wall has a 4" brick exterior, a 1" air gap, and 8" concrete block wall. Calculate the overall U value of the walls (Btu/hr/ft²/°F) based on the materials used and the standards set forth in ASHRAE. The tabulated values are as follows:

Construction Material	R-Value (hr x ft ² x °F/ Btu)
Outside Air Resistance	0.33
4" Face Brick	0.43
1" Air Gap	0.91
8" CMU	2.02
Inside Air Resistance	0.69
Totals	4.38

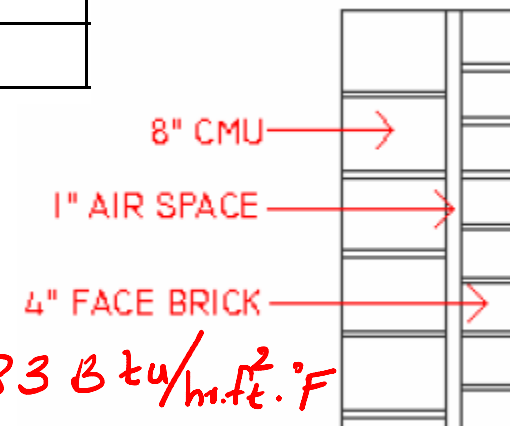


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$$\frac{1}{U} = \sum_i R_i \Rightarrow U = \frac{1}{\sum_i R_i}$$



∴ Overall heat transfer coefficient, $U = \frac{1}{4.38} = 0.2283 \text{ Btu/hr.ft}^2.\text{°F}$

External Cooling Load by Solar Load Through Glass



- Solar load through glass has two components
 1. Conductive and
 2. Solar Transmission
- The absorbed and then conductive portion of the radiation through the windows is treated like the roof & walls where CLTD values for standard glazing are tabulated in ASHARE fundamentals handbook.
- For solar transmission, the cooling load is calculated by the cooling load SCL factor and shading coefficient (SC).
- The cooling load equations for glass are:
 - Conductive $Q_{\text{Glass Conductive}} = U * A * \text{CLTD}_{\text{Glass Corrected}}$
 - Solar Transmission $Q_{\text{Glass Solar}} = A * SC * SCL$
 - $Q_{\text{Conductive}}$ = Conductive load through the glass in Btu/hr
 - Q_{Solar} = Solar transmission load through the glass in Btu/hr
 - U = Thermal Transmittance for glass in Btu/ (h ft² F)
 - A = area of glass in ft²
 - CLTD = Cooling Load Temperature Difference for glass in °F
 - SC = Shading coefficient
 - SCL = Solar Cooling Load Factor

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